Notes: Baker, Stein, and Wurgler (QJE, 2003)

When Does the Market Matter? Stock Prices and the Investment of Equity-Dependent Firms

Contents

1 Big picture

This note summarizes Baker, Stein, and Wurgler (2003), which asks when stock-market prices affect real corporate investment rather than merely forecasting investment fundamentals. The paper distinguishes a traditional Tobin's Q view from an equity-financing-channel view. Under the traditional view, stock prices matter because they summarize expected marginal product of capital. Under the equity financing channel, stock prices matter because mispricing changes the effective cost of external equity and therefore changes investment for firms that need external equity to finance marginal projects.

One-sentence summary. The paper develops a model in which nonfundamental stock-price movements affect investment only for equity-dependent firms, measures equity dependence using a modified Kaplan-Zingales index, and shows in Compustat data from 1980-1999 that high-equity-dependence firms have investment and equity issuance that are much more sensitive to Tobin's Q and future stock returns than low-equity-dependence firms.

Core empirical result. Firms in the top quintile of the modified KZ index have capital investment almost three times as sensitive to Q as firms in the bottom quintile: the investment-Q coefficient rises from about 0.012 in quintile 1 to about 0.033 in quintile 5. The same pattern appears in robustness tests, in alternative investment definitions, and when future realized returns are used as a noisy proxy for mispricing.

Main conceptual takeaway. The stock market matters most when a firm must issue equity at the margin. A cash-rich or financially unconstrained firm can ignore irrational stock-price movements. An equity-dependent firm cannot: if its shares are undervalued, issuing equity is costly, so investment falls; if its shares are overvalued, equity issuance becomes attractive and can support investment.

2 Incremental contribution of the paper

First, the paper makes the equity-financing channel empirically testable. Earlier work debated whether stock prices forecast investment after controlling for fundamentals, but it was difficult to distinguish fundamentals from mispricing. This paper adds a sharp cross-sectional prediction: if mispricing affects investment through equity finance, the effect should be strongest for firms that need external equity.

Second, the paper connects behavioral finance to real investment. A large literature documents that equity issuance and long-run returns are related to market timing. Baker, Stein, and Wurgler move from financing patterns to real effects: market inefficiency can influence actual capital expenditures through the cost of equity finance.

Third, the paper uses the Kaplan-Zingales index in a new way. Kaplan and Zingales (1997) originally study financing constraints and cash-flow sensitivity. This paper uses a modified KZ index as a proxy for *equity dependence*: the need to issue external equity to finance marginal investments. It deliberately removes Q from the index to avoid mixing the proxy for dependence with the stock-price variable of interest.

Fourth, the paper adds a mispricing proxy based on future returns. Tobin's Q contains fundamentals, mispricing, and measurement error. The authors therefore test whether investment is negatively related to future stock returns, especially among equity-dependent firms. The logic is that overvalued stocks should have low subsequent returns as mispricing corrects, while undervalued stocks should have high subsequent returns.

Fifth, the paper triangulates investment and financing. The model predicts not only investment sensitivity but also equity issuance sensitivity. The empirical evidence shows that high-KZ firms issue more equity when Q is high and issue less when future returns are high, supporting the idea that financing is the channel linking stock prices to investment.

3 Research question and main answer

The paper asks: **When does the market matter for real corporate investment?**

A simple positive correlation between investment and stock prices is not enough to prove that the market has an independent causal role. Stock prices may simply reveal investment opportunities. The paper therefore asks whether the relation is stronger exactly where the financing-channel theory says it should be stronger: among firms that need external equity to fund marginal investment.

The answer is: **the market matters when firms are equity-dependent**. In the data, firms that appear most dependent on external equity have investment that is much more sensitive to stock prices. They also show stronger links between stock prices, future returns, and equity issuance.

4 Theoretical framework

4.1 Basic setup

The model is a simplified version of Stein (1996). A firm can invest K at time 0 and receive gross payoff $f(K)$ at time 1. The efficient-market discount rate is r, so the net present value is

$$\frac{f(K)}{1+r} - K. \tag{1}$$

The first-best investment level K^{fb} satisfies

$$\frac{f'(K^{fb})}{1+r} = 1. \tag{2}$$

The firm's equity may be mispriced by a percentage δ. If $\delta > 0$, equity is overvalued; if $\delta < 0$, equity is undervalued. Debt is assumed fairly priced. The firm can issue equity e, subject to

$$0 \leq e \leq e^{\max}. \tag{3}$$

The firm also faces a leverage constraint linking financing and investment:

$$e + W - K(1 - \bar{D}) \geq 0, \tag{4}$$

where W is internal wealth or untapped borrowing capacity and $\bar{D}$ is the debt capacity of new investment. The lower is W, and the lower is $\bar{D}$, the more likely the firm must issue equity to invest.

The firm solves

$$\max_{e,K} \frac{f(K)}{1+r} - K + \delta e, \tag{5}$$

subject to the equity-issuance and leverage constraints.

4.2 Equity dependence

The key definition is:

$$\text{Firm is equity-dependent if} \quad W < K^{fb}(1-\bar{D}). \tag{6}$$

This means that internal resources plus debt capacity are insufficient to finance the first-best level of investment, so the firm needs outside equity at the margin.

Economic meaning.

- A firm with large cash holdings and debt capacity can finance investment even if its stock is undervalued.
- A firm with low cash, high leverage, low debt capacity, and strong investment opportunities may have to issue equity.
- If such a firm is undervalued, issuing equity transfers value from existing shareholders to new shareholders, making investment less attractive.

4.3 Model predictions

The model generates three main predictions.

Hypothesis 1: investment and Q. Equity-dependent firms should have investment that is more sensitive to stock prices, measured by Tobin's Q, than non-equity-dependent firms:

$$\frac{\partial I}{\partial Q} \quad \text{is larger for equity-dependent firms.} \tag{7}$$

Hypothesis 2: investment and future returns. If high prices partly reflect overvaluation, future returns should be lower. Thus, investment should be negatively related to future returns, especially for equity-dependent firms:

$$\frac{\partial I}{\partial R_{t,t+3}} \quad \text{is more negative for equity-dependent firms.} \tag{8}$$

Hypothesis 3: financing and stock prices. Equity-dependent firms should issue more equity when stock prices are high and less when future returns are high:

$$\frac{\partial e}{\partial Q} > 0, \qquad \frac{\partial e}{\partial R_{t,t+3}} < 0. \tag{9}$$

5 Data and measurement

5.1 Sample

The sample is an unbalanced panel of Compustat firms from 1980 to 1999. The authors exclude financial firms and firm-years with book assets below \$10 million. The main sample has 52,101 firm-year observations, averaging about 2,605 observations per year.

5.2 Investment variables

The baseline investment measure is

$$\frac{\text{CAPX}_{it}}{A_{it-1}}, \tag{10}$$

where CAPX is capital expenditure and A is book assets.

The paper also studies broader investment measures:

$$\frac{\text{CAPX}_{it} + \text{RD}_{it}}{A_{it-1}}, \tag{11}$$

$$\frac{\text{CAPX}_{it} + \text{RD}_{it} + \text{SGA}_{it}}{A_{it-1}}, \tag{12}$$

$$\frac{\Delta A_{it}}{A_{it-1}}. \tag{13}$$

The broader measures matter because investment may be intangible rather than purely physical.

5.3 Financing variables

Equity issuance is measured as

$$\frac{e_{it}}{A_{it-1}}, \tag{14}$$

where external equity issues are constructed from changes in book equity net of changes in retained earnings.

Total external finance is

$$\frac{e_{it} + d_{it}}{A_{it-1}}, \tag{15}$$

where d_{it} captures debt issuance.

5.4 Modified KZ index

The paper measures equity dependence with a modified Kaplan-Zingales index. The original KZ index includes Q, but the paper removes Q because Q is the stock-price variable being tested. The four-variable index is

$$\text{KZ}_{it} = -1.002\frac{\text{CF}_{it}}{A_{it-1}} - 39.368\frac{\text{DIV}_{it}}{A_{it-1}} - 1.315\frac{C_{it}}{A_{it-1}} + 3.139\text{LEV}_{it}. \tag{16}$$

Higher values of KZ imply greater equity dependence:

- low cash flow increases equity dependence;
- low dividends increase equity dependence;
- low cash balances increase equity dependence;
- high leverage increases equity dependence.

The economic interpretation is that high-KZ firms have fewer internal funds and less room to borrow, so they are more likely to need external equity.

5.5 Stock-price variables

The main stock-price variable is Tobin's Q, defined as market value of equity plus assets minus book equity, divided by assets. Q is used as a broad proxy for market valuation.

The second stock-price variable is the cumulative future stock return over years $t+1$, $t+2$, and $t+3$, denoted $R_{it,t+3}$. Future returns are used as a noisy proxy for mispricing: overvalued firms should tend to earn low future returns, and undervalued firms should tend to earn high future returns.

6 Empirical strategy

6.1 Investment and Q

The baseline regression is estimated separately by KZ quintile:

$$\frac{\mathrm{CAPX}_{it}}{A_{it-1}} = a_i + a_t + bQ_{it-1} + c\frac{\mathrm{CF}_{it}}{A_{it-1}} + u_{it}. \tag{17}$$

The key object is whether b rises with the KZ quintile. A rising pattern means that investment becomes more sensitive to stock prices as firms become more equity-dependent.

6.2 Interactive investment specification

The paper also estimates pooled specifications such as

$$\frac{I_{it}}{A_{it-1}} = a_i + a_t + bQ_{it-1} + c(Q_{it-1} \times \mathrm{KZ}_i) + d\frac{\mathrm{CF}_{it}}{A_{it-1}} + u_{it}. \tag{18}$$

The key coefficient is c. A positive c means that higher equity dependence amplifies the investment response to stock prices.

6.3 Future-return specification

To test the mispricing channel more directly, the paper replaces Q with future returns:

$$\frac{\mathrm{CAPX}_{it}}{A_{it-1}} = a_t + bR_{it,t+3} + c\frac{\mathrm{CF}_{it}}{A_{it-1}} + u_{it}. \tag{19}$$

The prediction is that b should be negative and more negative for high-KZ firms.

6.4 Financing specification

Finally, the paper uses the same logic with financing as the dependent variable:

$$\frac{e_{it}}{A_{it-1}} \quad \text{or} \quad \frac{e_{it}+d_{it}}{A_{it-1}}. \tag{20}$$

If the mechanism is truly an equity-financing channel, then high-KZ firms should issue more equity when stock prices are high and less when future returns are high.

7 Main empirical findings

7.1 Finding 1: investment-Q sensitivity rises strongly with equity dependence

The baseline estimates show that the coefficient on Q rises from 0.012 in the lowest KZ quintile to 0.033 in the highest KZ quintile. This means that the investment of the most equity-dependent firms is almost three times as sensitive to stock prices as the investment of the least equity-dependent firms.

7.2 Finding 2: the result is robust

The pattern survives many robustness checks: using the original five-variable KZ index, using five-year median or annual KZ rankings, scaling KZ components by PP&E instead of assets, adding lagged investment, excluding cash flow, restricting to low-dividend manufacturing firms, using the complementary sample, and resetting the index weights.

7.3 Finding 3: broader investment measures show the same pattern

When investment is expanded to include R&D, selling/general/administrative expenses, or total asset growth, the interaction between Q and the KZ index remains positive and significant. This matters because equity-dependent firms may invest in intangible assets, not only physical capital.

7.4 Finding 4: KZ components behave as the theory predicts

The investment-Q sensitivity is stronger for young firms, non-dividend-paying firms, low-cash firms, low-cash-flow firms, high-leverage firms, and firms in industries with volatile cash flows. This detailed pattern fits the theory's definition of equity dependence.

7.5 Finding 5: future returns support the mispricing interpretation

Investment is negatively related to future returns, and this negative sensitivity is stronger for high-KZ firms. This is consistent with the idea that managers invest more when their stock appears overvalued and subsequent returns are low. It is harder to explain this result using stories based only on measurement error in Q or cross-firm differences in production technologies.

7.6 Finding 6: financing patterns support the channel

High-KZ firms issue more equity when Q is high and issue less equity when future returns are high. This is exactly what the equity-financing-channel mechanism requires. The results also show that total external finance responds strongly, implying that equity-dependent firms raise equity and debt together when market conditions are favorable.

8 Interpretation of tables and figures

8.1 Figure I: Graphical Illustration of Proposition 1

Read. Figure I shows how mispricing affects investment and equity issuance differently for equity-dependent and non-equity-dependent firms. When equity is overvalued, both types of firms can invest at the first-best level and issue equity. When equity is undervalued, only equity-dependent firms cut investment because they would have to issue undervalued shares to finance the marginal project.

Economic message. The figure is the model's central intuition: mispricing does not affect every firm equally. It matters when the firm must rely on the equity market. This is the theoretical reason the empirical tests focus on KZ-sorted firms.

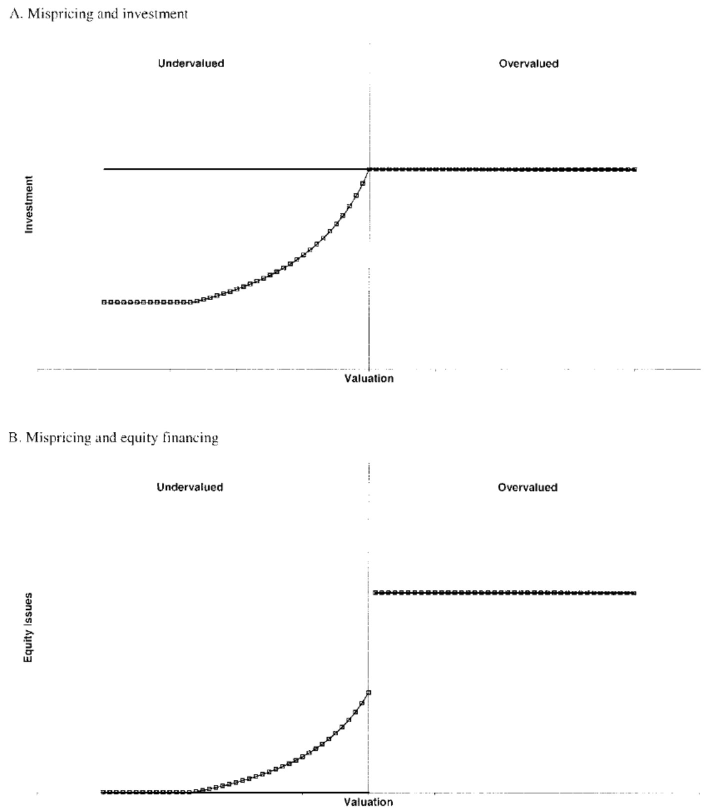

FIGURE I

Graphical Illustration of Proposition 1

The horizontal axis measures the degree of stock-price misvaluation δ. The vertical axis shows investment K (in Panel A) and equity financing e (in Panel B). The square markers show the policy of an equity-dependent firm. The solid line is the policy of a nonequity-dependent firm.

Figure I: Graphical Illustration of Proposition 1

8.2 Table I: Summary Statistics for Investment and Financing Variables; Table II: Summary Statistics for Equity Dependence and Other Investment Determinants

Read. Table I reports the distribution of the main investment and financing variables. Baseline capital expenditures average 8.23 percent of assets, while broader measures including R&D and SG&A are much larger. Equity issuance averages 4.38 percent of assets and total external finance averages 10.51 percent. Table II reports the ingredients of equity dependence and other investment determinants, including the KZ index, cash flow, dividends, cash, leverage, age, industry cash-flow volatility, Q, future returns, and cash flow.

Economic message. The data contain substantial variation in both real investment and external financing. The KZ index combines low cash flow, low dividends, low cash, and high leverage into a practical proxy for firms that likely need external equity.

TABLE I

SUMMARY STATISTICS FOR INVESTMENT AND FINANCING VARIABLES

	Full sample						Subsample means	
	N	Mean	SD	Median	Min	Max	1980–1989	1990–1999
Panel A: Investment								
$CAPX_t/A_{t-1}\%$	52,101	8.23	7.85	6.00	0.18	45.22	8.81	7.79
$+RD_t/A_{t-1}$	52,101	11.41	10.46	8.45	0.24	58.43	10.96	11.76
$+RD_t$ $+ SGA_t/A_{t-1}$	52,101	40.09	28.93	34.21	1.65	144.04	39.23	40.76
$\Delta A_t/A_{t-1}\%$	52,101	11.10	28.03	6.41	−43.76	153.73	10.30	11.73
Panel B: Financing								
$e_t/A_{t-1}\%$	52,101	4.38	14.48	0.64	−15.90	92.28	3.00	5.44
$+d_t/A_{t-1}$	52,101	10.51	27.32	4.03	−33.70	161.59	8.48	12.08

Source: Compustat, 1980–1999.

a. Investment is alternately defined as capital expenditures (Item 128) over assets (Item 6); capital expenditures plus research and development expenses (Item 46) over assets; capital expenditures plus research and development expenses plus selling, general, and administrative expenses (Item 189) over assets; and growth in assets.

b. Financing is alternately defined as equity issues (ΔItem 60 + ΔItem 74 − ΔItem 36) (i.e., the change in book equity minus the change in retained earnings) over assets, and equity issues plus debt issues (ΔItem 6 − ΔItem 60 − ΔItem 74) over assets.

c. All variables are Winsorized at the first and ninety-ninth percentiles.

(a) Table I: Summary Statistics for Investment and Financing Variables

TABLE II

SUMMARY STATISTICS FOR EQUITY DEPENDENCE AND OTHER INVESTMENT DETERMINANTS

	Full sample						Subsample means	
	N	Mean	SD	Median	Min	Max	1980–1989	1990–1999
Panel A: Equity dependence variables $(t-2)$								
KZ Index	52,101	−0.04	2.60	0.28	−18.43	3.68	−0.12	0.03
$CF_{t-2}/A_{t-3}\%$	52,101	12.57	28.75	8.97	−39.81	225.92	13.31	11.99
$DIV_{t-2}/A_{t-3}\%$	52,101	1.98	4.67	0.60	0.00	37.65	2.34	1.71
$C_{t-2}/A_{t-3}\%$	52,101	14.95	30.13	5.01	0.01	223.79	12.80	16.61
$LEV_{t-2}\%$	52,101	35.33	25.62	34.29	0.00	124.13	35.76	35.00
AGE_{t-2}	52,101	10.72	6.08	9.00	1.00	31.00	8.86	12.17
Industry $\sigma(CF/A)$	52,101	144.49	223.89	64.89	11.34	914.11	142.44	146.08
Panel B: Other investment determinants								
Q_{t-1}	52,101	1.47	0.93	1.17	0.53	6.15	1.28	1.62
$R_{it,t+3}\%$	41,819	58.37	111.70	36.96	−92.32	565.24	54.79	62.63
$CF_t/A_{t-1}\%$	52,101	8.20	11.68	9.19	−42.43	36.52	9.25	7.39

Source: Compustat and CRSP, 1980–1999.

a. Equity dependence is defined using the Kaplan and Zingales [1997] index of financial constraints (excluding Q from the index). This modified version of the index has four components: cash flow (Item 14 + Item 18) over assets; cash dividends (Item 21 + Item 19) over assets; cash balances (Item 1) over assets; and leverage ((Item 9 + Item 34)/(Item 9 + Item 34 + Item 216)). We also consider two additional measures of equity dependence, firm age and the industry standard deviation of cash flow over assets between 1980 and 1999. Industry definitions follow Fama and French [1997]. Age is equal to the current year minus the IPO year.

b. Q is defined as the market value of equity (price times shares outstanding from CRSP) plus assets minus the book value of equity (Item 60 + Item 74) over assets. $R_{it,t+3}$ is the cumulative stock return in years $t+1$, $t+2$ and $t+3$ from CRSP.

c. All variables are Winsorized at the first and ninety-ninth percentiles except for firm age and the industry standard deviation of cash flow.

(b) Table II: Summary Statistics for Equity Dependence and Other Investment Determinants

8.3 Table III: Equity Dependence and the Link between Investment and Stock Prices; Figure II: Equity Dependence and the Link between Investment and Stock Prices

Read. Table III presents the baseline investment regressions by KZ quintile. The Q coefficient rises monotonically from 0.012 in quintile 1 to 0.033 in quintile 5. Figure II shows the same relationship using twenty KZ groups rather than five: the sensitivity of investment to Q rises steadily with equity dependence.

Economic message. This is the paper's main evidence. If Q were only a fundamentals proxy, there would be no obvious reason for its investment coefficient to rise so sharply with equity dependence. The pattern is exactly what the equity-financing-channel model predicts.

TABLE III

EQUITY DEPENDENCE AND THE LINK BETWEEN INVESTMENT AND STOCK PRICES

		Q_{t-1}			CF_t/A_{t-1}			
KZ index	N	b	(se)	[t-stat]	c	(se)	[t-stat]	R^2
Quintile 1	10,427	0.012	(0.0013)	—	0.110	(0.0146)	—	0.62
2	10,415	0.018	(0.0015)	[2.86]	0.124	(0.0133)	[0.74]	0.61
3	10,421	0.024	(0.0024)	[4.45]	0.117	(0.0140)	[0.38]	0.58
4	10,418	0.032	(0.0030)	[6.17]	0.137	(0.0130)	[1.42]	0.62
5	10,420	0.033	(0.0042)	[4.80]	0.145	(0.0134)	[1.80]	0.62

a. Regressions of investment on Q and cash flow by equity dependence quintile. We sort firms into five quintiles according to the firm median Kaplan and Zingales [1997] index of financial constraints (excluding Q from the index), performing separate regressions for each group. Year and firm fixed effects are included.
b. Investment is defined as capital expenditures over assets. Q is defined as the market value of equity plus assets minus the book value of equity over assets. Cash flow is defined as operating cash flow over assets.
c. t-statistics test the hypothesis of no difference between the coefficient in each quintile and quintile 1.
d. Standard errors and t-statistics are heteroskedasticity-robust, clustered by firm, with all five regressions estimated simultaneously.

(a) Table III: Equity Dependence and the Link between Investment and Stock Prices

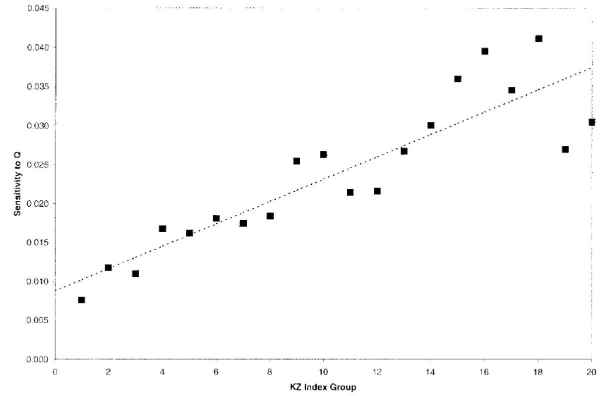

FIGURE II

Equity Dependence and the Link between Investment and Stock Prices

The figure shows the sensitivity of investment to Q by equity dependence group. We sort firms into twenty groups according to the firm median Kaplan and Zingales [1997] index of financial constraints (excluding Q from the index) over

(b) Figure II: Equity Dependence and the Link between Investment and Stock Prices

8.4 Table IV: Equity Dependence and the Link between Investment and Stock Prices - Robustness; Table V: Equity Dependence and the Link between Investment and Stock Prices - Interactive Specifications

Read. Table IV shows that the main result is robust to many changes in the KZ index and regression design. The top-minus-bottom coefficient difference remains positive and statistically significant in every row. Table V uses pooled interaction specifications and alternative investment measures. The coefficient on $Q \times$ KZ is positive and significant for capital expenditures, capital expenditures plus R&D, capital expenditures plus R&D plus SG&A, and asset growth.

Economic message. The result is not an artifact of a particular way of constructing KZ, scaling variables, or measuring investment. The stock market matters more for equity-dependent firms across many reasonable empirical choices.

TABLE IV

EQUITY DEPENDENCE AND THE LINK BETWEEN INVESTMENT AND STOCK PRICES: ROBUSTNESS

	Bottom quintile		2		3		4		Top quintile		Top − bottom	
	b	(se)	b	(se)	b	(se)	b	(se)	b	(se)	Δb	[t-stat]
Z (base case)	0.012	(0.0013)	0.018	(0.0015)	0.024	(0.0024)	0.032	(0.0030)	0.033	(0.0042)	0.021	[4.80]
Z (five-variable index)	0.011	(0.0015)	0.019	(0.0017)	0.020	(0.0020)	0.028	(0.0024)	0.027	(0.0030)	0.016	[4.74]
Z (five-year median)	0.011	(0.0014)	0.019	(0.0017)	0.024	(0.0022)	0.030	(0.0029)	0.034	(0.0046)	0.023	[4.72]
Z (annual)	0.010	(0.0016)	0.019	(0.0019)	0.024	(0.0029)	0.033	(0.0036)	0.035	(0.0045)	0.026	[5.40]
Z (PP&E scaling)	0.010	(0.0012)	0.015	(0.0014)	0.025	(0.0021)	0.032	(0.0030)	0.039	(0.0039)	0.030	[7.28]
gged *CAPX/A* included	0.009	(0.0012)	0.012	(0.0014)	0.019	(0.0021)	0.025	(0.0027)	0.028	(0.0037)	0.019	[4.96]
F/A excluded	0.016	(0.0013)	0.022	(0.0016)	0.028	(0.0025)	0.037	(0.0033)	0.038	(0.0048)	0.022	[4.47]
w-div. mfg., PP&E scaling	0.008	(0.0012)	0.013	(0.0019)	0.015	(0.0023)	0.025	(0.0033)	0.024	(0.0040)	0.016	[3.69]
mplement sample to #8	0.018	(0.0026)	0.024	(0.0031)	0.028	(0.0043)	0.042	(0.0045)	0.047	(0.0068)	0.030	[4.12]
Z (reset to equal weights)	0.014	(0.0012)	0.020	(0.0018)	0.027	(0.0027)	0.035	(0.0035)	0.031	(0.0047)	0.017	[3.60]

Regressions of investment on Q and cash flow by equity dependence quintile. We sort firms into five quintiles according to various modifications of the Kaplan and Zingales index of financial constraints, performing separate regressions for each group. The cash-flow coefficients are not reported. Year and firm fixed effects are included.
Investment is defined as capital expenditures over assets. Q is defined as the market value of equity plus assets minus the book value of equity over assets. Cash flow is defined rating cash flow over assets.
The first row is our baseline specification, which classifies firms according to their median four-variable KZ index (excluding Q). The second row uses the firm median riable KZ index (including Q). The third row uses a five-year median KZ index (excluding Q). The fourth row uses an annual KZ index measured at time $t-2$ (excluding Q). th row scales the KZ-index components by fixed assets instead of total assets. The sixth row includes lagged investment as an independent variable. The seventh row excludes ow. The eighth row restricts the sample to manufacturing firms (SIC codes 2000 through 3999) with a ratio of dividends to net income of less than 10 percent, and also scales -index components by fixed assets. The ninth row studies the complement sample to that studied in the eighth row, again scaling by fixed assets. The tenth row resets the lex weights so that each variable explains the same portion of the index variation.
Standard errors and t-statistics are heteroskedasticity-robust, clustered by firm, with all five regressions in a row estimated simultaneously.

(a) Table IV: Robustness

TABLE V

EQUITY DEPENDENCE AND THE LINK BETWEEN INVESTMENT AND STOCK PRICES: INTERACTIVE SPECIFICATIONS

		Q_{t-1}		$Q_{t-1} \cdot KZ$		CF_t/A_{t-1}		
	N	b	[t-stat]	c	[t-stat]	d	[t-stat]	R^2
CAPX/A	52,101	0.021	[22.04]	0.011	[6.87]	0.129	[20.96]	0.61
+RD/A	52,101	0.028	[21.63]	0.013	[6.71]	0.109	[12.89]	0.69
+RD + SGA/A	52,101	0.049	[18.01]	0.021	[4.19]	0.258	[14.64]	0.84
ΔA/A	52,101	0.080	[20.43]	0.058	[10.16]	1.117	[38.00]	0.41

a. Regressions of investment on Q, Q interacted with equity dependence, and cash flow. Year and firm fixed effects are included.
b. Investment is alternately defined as capital expenditures over assets; capital expenditures plus research and development expenses over assets; capital expenditures plus research and development expenses plus selling, general, and administrative expenses over assets; and growth in assets. Q is defined as the market value of equity plus assets minus the book value of equity over assets.
c. The measure of equity dependence is the firm median Kaplan and Zingales [1997] index of financial constraints (excluding Q from the index), standardized to have unit variance.
d. Cash flow is defined as operating cash flow over assets.

(b) Table V: Interactive Specifications

8.5 Table VI: Equity Dependence Decomposition; Table VII: Equity Dependence and the Link between Investment and Future Returns

Read. Table VI decomposes the KZ index into its components and adds firm age and industry cash-flow volatility. The investment-Q sensitivity is stronger for firms with characteristics associated with equity dependence: low dividends, low cash, high leverage, young age, and high industry cash-flow volatility. Table VII replaces Q with future stock returns. The future-return coefficient becomes more negative moving from low-KZ to high-KZ firms.

Economic message. Table VI shows that the KZ result has a coherent economic structure rather than being a black-box index result. Table VII strengthens the mispricing interpretation because future returns are a different proxy for nonfundamental valuation than Q.

TABLE VI

EQUITY DEPENDENCE DECOMPOSITION

	CF/A		DIV/A		C/A		LEV		$\log(AGE_t)$		Industry $\sigma(CF/A)$	
	c_1	[t-stat]	c_2	[t-stat]	c_3	[t-stat]	c_4	[t-stat]	c_5	[t-stat]	c_6	[t-stat]
index	−		−		−		+					
liction	−		−		−		+		−		+	
Panel A: KZ index variables												
X/A	0.006	[3.68]	−0.006	[−4.07]	−0.006	[−5.66]	0.007	[4.79]				
RD/A	−0.000	[−0.03]	−0.006	[−3.36]	−0.004	[−2.47]	0.008	[4.14]				
RD + SGA/A	−0.020	[−3.69]	0.003	[0.59]	0.002	[0.68]	0.022	[5.92]				
A	−0.025	[−2.92]	−0.030	[−6.17]	0.002	[0.34]	0.027	[4.93]				
Panel B: Additional measures of equity dependence												
X/A	0.006	[4.16]	−0.004	[−3.31]	−0.006	[−6.26]	0.005	[3.43]	−0.004	[−5.67]	0.007	[4.93]
RD/A	0.000	[0.17]	−0.005	[−2.73]	−0.004	[−2.85]	0.006	[3.08]	−0.004	[−3.98]	0.007	[4.14]
RD + SGA/A	−0.019	[−3.47]	0.006	[1.22]	0.000	[0.10]	0.022	[5.34]	−0.008	[−3.65]	0.004	[1.67]
A	−0.024	[−2.84]	−0.026	[−5.35]	0.000	[0.00]	0.025	[4.23]	−0.011	[−4.02]	0.010	[2.77]

. Regressions of investment on Q, Q interacted with the components of equity dependence, and cash flow. Only the coefficients on the Q interaction terms are reported. Year irm fixed effects are included.

. Investment is alternately defined as capital expenditures over assets; capital expenditures plus research and development expenses over assets; capital expenditures plus rch and development expenses plus selling, general, and administrative expenses over assets; and growth in assets. Q is defined as the market value of equity plus assets minus ook value of equity over assets.

. The first panel decomposes the effect of the Kaplan and Zingales [1997] index (excluding Q) into its four components: firm median operating cash flow over assets; firm median ends over assets; firm median cash balance over assets; and firm median leverage. The second panel adds two additional measures of equity dependence, firm age, and the stry standard deviation of cash flow over assets between 1980 and 1999. The level of firm age is included along with the interaction between firm age and Q.

. All components of equity dependence are standardized to have unit variance.

. t-statistics use heteroskedasticity-robust standard errors, clustered by firm.

(a) Table VI: Equity Dependence Decomposition

TABLE VII

EQUITY DEPENDENCE AND THE LINK BETWEEN INVESTMENT AND FUTURE RETURNS

		$R_{it,t+3}$			CF_t/A_{t-1}			
KZ index	N	b	(se)	[t-stat]	c	(se)	[t-stat]	R^2
Quintile 1	8307	−0.004	(0.0009)	—	0.228	(0.0169)	—	0.14
2	8292	−0.004	(0.0011)	[−0.33]	0.215	(0.0165)	[−0.57]	0.07
3	8449	−0.006	(0.0010)	[−1.66]	0.250	(0.0207)	[0.82]	0.07
4	8407	−0.008	(0.0009)	[−3.08]	0.311	(0.0204)	[3.11]	0.08
5	8364	−0.007	(0.0009)	[−2.40]	0.290	(0.0186)	[2.44]	0.08

a. Regressions of investment on future returns and cash flow by equity dependence quintile. We sort firms into five quintiles according to the firm median Kaplan and Zingales [1997] index of financial constraints (excluding Q from the index), performing separate regressions for each group. Year fixed effects are included.

b. Investment is defined as capital expenditures over assets. $R_{it,t+3}$ is the cumulative stock return in years $t+1$, $t+2$, and $t+3$ from CRSP. Cash flow is defined as operating cash flow over assets.

c. t-statistics test the hypothesis of no difference between the coefficient in each quintile and quintile 1.

d. Standard errors and t-statistics are heteroskedasticity-robust, clustered by firm, with all five regressions in a panel estimated simultaneously.

(b) Table VII: Equity Dependence and the Link between Investment and Future Returns

8.6 Figure III: Equity Dependence and the Link between Investment and Future Returns; Table VIII: Equity Dependence and the Link between External Finance, Stock Prices, and Returns

Read. Figure III plots the investment-future-return sensitivity across twenty KZ groups. The slope is downward: high-equity-dependence firms have more negative investment sensitivity to future returns. Table VIII turns to financing. Equity issuance is strongly positively related to Q, especially for high-KZ firms, and total external finance shows an even stronger pattern. Future returns are negatively related to financing, again especially in the total external finance measure.

Economic message. Figure III supports the idea that high stock prices are not merely forecasting fundamentals: if they predict low future returns and high current investment among equity-dependent firms, the pattern is consistent with market timing and mispricing. Table VIII completes the mechanism by showing that the firms whose investment is most stock-price-sensitive are also the firms whose external financing is most stock-price-sensitive.

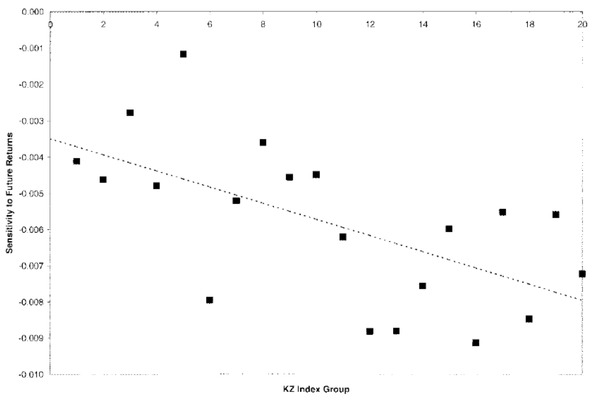

FIGURE III

Equity Dependence and the Link between Investment and Future Returns

The figure shows the sensitivity of investment to future stock returns by equity dependence group. We sort firms into twenty groups according to the firm median Kaplan and Zingales [1997] index of financial constraints (excluding Q from the index) over the period from 1980 to 1999, performing separate regressions for each

(a) Figure III: Equity Dependence and the Link between Investment and Future Returns

TABLE VIII

EQUITY DEPENDENCE AND THE LINK BETWEEN EXTERNAL FINANCE, STOCK PRICES, AND RETURNS

	Bottom quintile		2		3		4		Top quintile		Top − bottom	
	b	(se)	b	(se)	b	(se)	b	(se)	b	(se)	Δb	[t-stat]
Panel A: Financing and Q												
rnal equity—all firms	0.021	(0.0040)	0.051	(0.0050)	0.040	(0.0056)	0.056	(0.0069)	0.064	(0.0085)	0.043	[4.59]
rnal equity plus debt—all	0.036	(0.0065)	0.077	(0.0080)	0.087	(0.0090)	0.104	(0.0115)	0.136	(0.0161)	0.100	[5.75]
Panel B: Financing and future returns												
rnal equity—all firms	−0.007	(0.0018)	−0.013	(0.0021)	−0.012	(0.0020)	−0.012	(0.0012)	−0.008	(0.0010)	−0.001	[−0.71]
rnal equity plus debt—all	−0.008	(0.0030)	−0.019	(0.0031)	−0.020	(0.0033)	−0.026	(0.0025)	−0.020	(0.0026)	−0.012	[−2.94]

Regressions of external finance on Q and cash flow, by equity dependence quintile (Panel A), and regressions of external finance on future returns and cash flow, by equity lence quintile (Panel B). We sort firms into five quintiles according to the firm median Kaplan and Zingales [1997] index of financial constraints (excluding Q), performing te regressions for each group.
External financing is alternately defined as equity issues over assets and as equity plus debt issues over assets. Q is defined as the market value of equity plus assets minus ok value of equity over assets. $R_{it,t+3}$ is the cumulative stock return in years $t+1$, $t+2$, and $t+3$ from CRSP. Cash flow is defined as operating cash flow over assets.
Year fixed effects are included in both panels. Firm fixed effects are included in the first panel only.
Standard errors and t-statistics are heteroskedasticity-robust, clustered by firm, with all five regressions in a row estimated simultaneously.

(b) Table VIII: External Finance, Stock Prices, and Returns

9 How to understand the identification

9.1 What the paper can identify

The paper does not claim to directly observe mispricing. Instead, it uses cross-sectional comparative statics. The theoretical mechanism predicts that stock prices should matter more for firms dependent on external equity. The empirical design tests exactly this: it compares the stock-price sensitivity of investment across firms with different degrees of equity dependence.

The strongest evidence comes from the combination of three facts:

1. investment-Q sensitivity rises with the KZ index;
2. investment-future-return sensitivity becomes more negative with the KZ index;
3. equity issuance sensitivity to Q and future returns is strongest for high-KZ firms.

Together, these facts point to a financing mechanism rather than a purely technological or measurement-error story.

9.2 Why future returns matter

Tobin's Q can reflect fundamentals, mispricing, or measurement error. Future realized returns help because they are related to expected returns and potential correction of mispricing. If high-valuation firms subsequently earn low returns, this is consistent with overvaluation. Therefore, a negative relationship between current investment and future returns, especially for equity-dependent firms, supports the mispricing channel.

9.3 Why financing evidence matters

The model is not only about investment. It is about investment funded through equity issuance. Therefore, the paper needs to show that firms identified as equity-dependent actually raise equity when stock prices are high. Table VIII provides this link. Without financing evidence, the investment results could be interpreted as a reduced-form correlation. With financing evidence, the channel becomes more credible.

10 Relation to alternative explanations

10.1 Fundamentals explanation

A standard Tobin's Q model says that investment responds to stock prices because stock prices summarize expected profitability. The paper does not deny this. Instead, it asks whether the response is stronger for equity-dependent firms. A pure fundamentals story must explain why Q is a much stronger investment predictor precisely among firms that need external equity.

10.2 Adverse selection explanation

Another possibility is that low-Q firms face worse adverse selection in equity markets, making investment more sensitive to Q. But this efficient-market adverse-selection story does not naturally predict the future-return results. If investment is especially high when future returns are low among high-KZ firms, this is more consistent with mispricing than with adverse selection alone.

10.3 Measurement error in Q

Measurement error in Q could attenuate investment-Q coefficients. But for this to explain the results, measurement error would need to be systematically lower for high-KZ firms. This is the opposite of common concerns in the financial constraints literature, where young, intangible, high-growth firms may have noisier Q. The detailed KZ decomposition and future-return tests make a simple measurement-error story less plausible.

11 Conceptual contribution for finance and macro-finance

This paper is important because it links asset prices to real activity through financing frictions. In a frictionless Modigliani-Miller world, mispricing should not affect real investment: firms can finance projects independently of market sentiment. In this paper's world, external equity is costly when shares are undervalued, so investment becomes sensitive to market mispricing.

The mechanism is especially useful for thinking about episodes such as technology booms, IPO waves, and sectoral investment booms. When equity markets assign high valuations to a set of firms, those firms can raise equity cheaply. If they are equity-dependent, this may generate real investment effects.

12 Limitations and caveats

First, mispricing is not directly observed. The paper uses Q and future returns as proxies. This is powerful but imperfect. Q combines fundamentals and mispricing; future returns are noisy realized outcomes.

Second, KZ is a proxy, not a structural measure. The modified KZ index captures plausible dimensions of equity dependence, but it does not directly observe a firm's shadow cost of equity finance.

Third, welfare conclusions are subtle. The paper warns that greater sensitivity of investment to stock prices is not automatically inefficient. If unconstrained managers smooth investment too much or ignore useful market information, then stock-price sensitivity could sometimes improve investment efficiency.

Fourth, the empirical setting is mostly US public firms. The mechanism may be stronger or weaker in private firms, firms in bank-based systems, developing economies, or periods with different equity issuance costs.

13 Short summary

Baker, Stein, and Wurgler (2003) argue that stock prices matter for real investment when firms depend on external equity. Their model predicts that market mispricing affects investment only for firms that must issue equity to fund marginal investment. Using a modified Kaplan-Zingales index to proxy for equity dependence, they show that high-KZ firms have investment almost three times as sensitive to Q as low-KZ firms. They also show that high-KZ firms' investment is more negatively related to future returns and that their equity issuance responds more strongly to valuation. The paper therefore provides evidence that market inefficiency can have real effects through an equity financing channel.

14 Conclusion

Core conclusion. The market matters when firms need the market. Stock prices influence real investment most strongly among equity-dependent firms, where issuing equity is necessary to finance marginal projects.

Economic intuition. A financially unconstrained firm can ignore irrational stock-price fluctuations. An equity-dependent firm cannot. When its stock is undervalued, the cost of issuing equity rises, discouraging investment. When valuation is high, equity finance becomes easier, increasing investment and issuance.

Incremental contribution revisited. The paper contributes by turning a broad question – whether the stock market affects investment – into a cross-sectional prediction about which firms should be affected. This makes the equity-financing channel empirically testable.

Final takeaway. Stock prices are not just passive signals of fundamentals. For firms that rely on external equity, valuation can shape real investment.